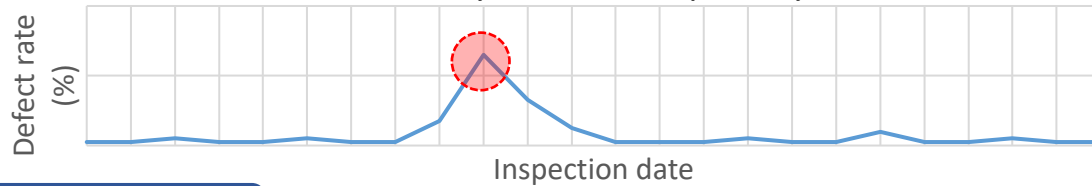


Identified and Resolved an ESD Defect Spike via Dielectric Thickness Control

Problem

ESD defect rate spike

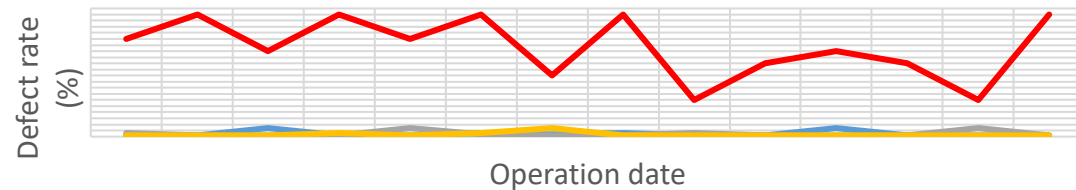
- A sudden spike in the electrostatic discharge (ESD) defect rate occurred
- The defects caused reliability failures, scrap, and yield loss



Investigation

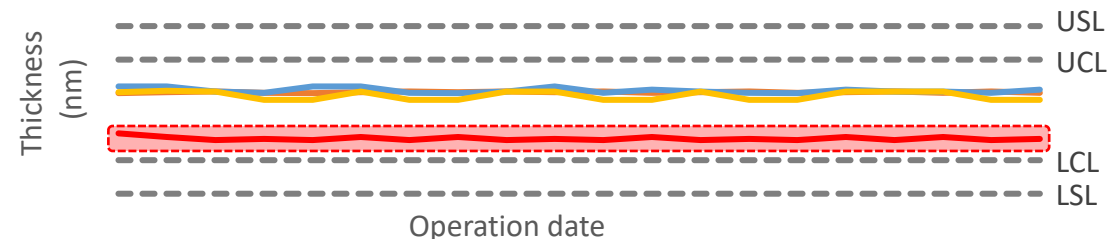
Tool-to-tool defect rate variation

- **One photolithography coater** showed a higher defect rate than others
- Confirmed through tool-level analysis at a specific process step



Lower thickness

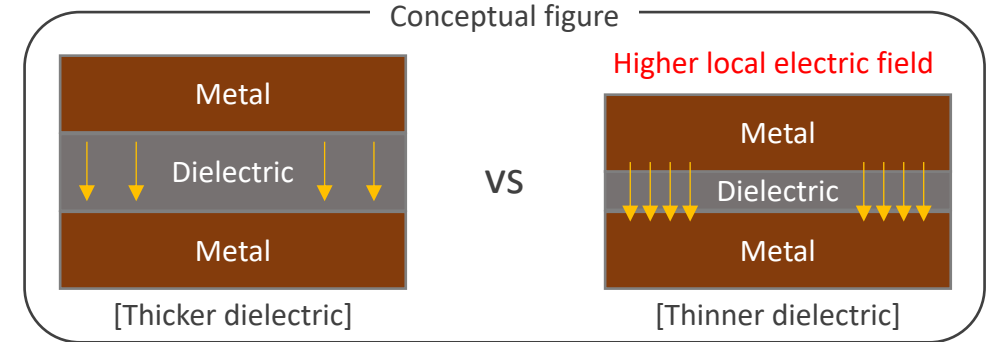
- The same photolithography coater produced a **thinner layer**
- Confirmed by inline thickness monitoring data and SEM analysis



Hypothesis

Capacitive coupling effect

- **Thinner dielectrics** in the capacitive stack led to **higher local electric fields**, increasing ESD susceptibility



Verification

Higher thickness target

- Resulted in a **lower defect rate** in tool-level pre/post comparisons

FIB-SEM analysis

- Showed that most ESD-defective samples had **thinner dielectric layers**

Action

Thickness spec & control limit

- **Adjusted the thickness target** and **redefined spec** with related teams

Thickness monitoring

- Implemented automated thickness monitoring with email alerts
- Increased thickness measurement frequency